VEGETATION–CLIMATE PRINCIPAL COMPONENT ANALYSIS

Environmental Data Cube for Oglala Lakota

Ate Makoce | May–September 2000–2023 | 24 annual growing seasons

Annual PCA scores and environmental correlation loadings

RESULTS SUMMARY

PC1 explained 81.95% of standardized variance; PC2 explained 12.11% (94.06% combined).

Positive PC1 corresponds to higher vegetation indices and precipitation, and lower maximum temperature.

Arrows show variable correlations with PC1 and PC2, rescaled for display. Associations do not establish causation or grazing impacts.